

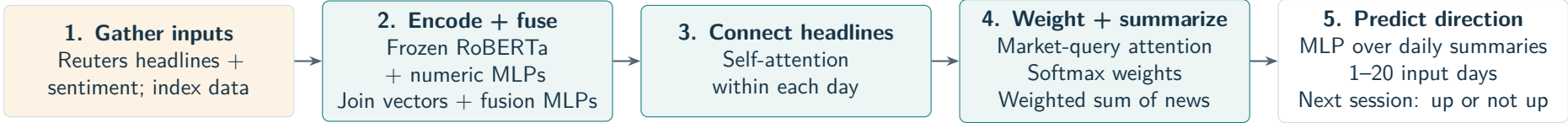
From the paper's idea to our prototype

Goal: use financial news and market data to predict whether the next trading session closes higher.

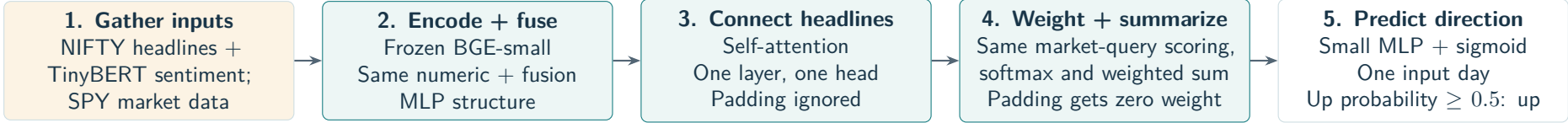
The key idea: first read headlines **in relation to each other**, then use the market situation to decide **how much weight each headline receives**.

Same core pipeline, smaller inputs

THE PAPER



OUR PROTOTYPE



How the blocks work: text shares an encoder/projection; numbers use separate MLPs. Fusion joins text and numbers for each headline and for the market. The market queries the updated news; softmax turns scores into weights, then a weighted sum creates one news summary. Market features also feed the predictor.

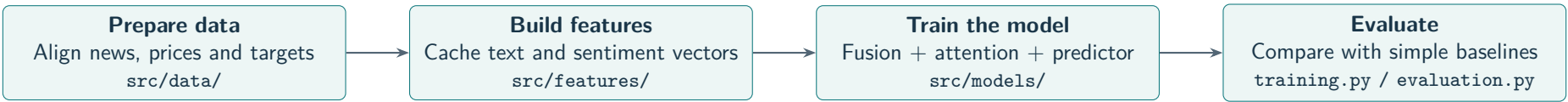
The essential differences

	Paper	Our prototype
Data	S&P 500 + NASDAQ 100; 2003–2018	SPY (S&P 500 ETF proxy); 2010–2020
News	Over 2.7 million Reuters items	2,107 usable dates; at most 16 NIFTY headlines/date
Text / sentiment	RoBERTa in main results; Reuters sentiment	BGE-small text vectors; TinyBERT sentiment
Experiment	1–20 input days; ten time windows	One input day; one time split; three training seeds

MLP = a small feed-forward neural network. **Softmax** = weights that sum to one. The lookback changes; the forecast stays **one session ahead**.

What is working, and what have we learned?

The prototype runs from data to saved predictions



Frozen: pretrained BGE-small and TinyBERT. **Trained:** text projection, numeric/fusion MLPs, attention and predictor.

One prediction, explained simply

6 Jan: news → **7 Jan: forecast** → **8 Jan: answer**

Use the news from 6 January and prices through the close of 7 January. Predict whether 8 January closes higher.

We delay news by one trading session because individual publication times are missing. Future prices never enter the model’s inputs.

Verified, not just implemented

2,107 examples rebuilt and checked.

18 training runs reproduce saved predictions.

7 integrity tests pass.

Data split: 1,475 train / 315 validation / 317 test.

The result: a working prototype, with no clear forecasting gain yet

On the same 317 test dates	Correct predictions	Balanced accuracy*
Always predict up	58.0%	50.0%
Our reduced FININ	58.4%	50.4%

*Gives equal importance to up and non-up days. Prototype values are averages across three seeds.

Why the scores are close: the model predicts up on 312–317 of the 317 dates. It mostly follows the majority class. The experiment does **not yet show that news attention improves prediction**.

Next step: investigate this behavior on training/validation data, then test longer news histories against the same simple baselines using a fresh temporal evaluation.

Sources: Wang, Cohen & Ma (2024), *Modeling News Interactions and Influence for Financial Market Prediction*, Figure 2 and Sections 3–5; [NIFTY dataset](#); local experiments and audit. This is an architectural prototype; the paper’s published gains have not been reproduced.